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Thesis title

Author One

student record book number 123456

Author Two

student record book number 654321

thesis supervisor

title/degree Name Surname

consultation (optional)

academic title/degree Name and Surname

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Abstract

Neural PK-PD Modeling with ODE: Multi-Task Molecular Property Prediction and Decision-Oriented Simulation

This thesis presents an end-to-end machine learning framework for early drug development that combines molecular property prediction with mechanistic pharmacokinetic-pharmacodynamic (PK-PD) simulation. The proposed workflow integrates heterogeneous public datasets, performs robust feature engineering and data quality checks, and trains multi-task neural models to predict binding affinity, hERG-related cardiotoxicity, Caco-2 permeability, and clearance.

The core contribution is a physics-informed decision pipeline where learned molecular endpoints are coupled with a Neural ODE-based PK-PD simulator for scenario analysis, sensitivity analysis, dose optimization, Pareto trade-off exploration, Monte Carlo validation, and in-silico clinical trial simulation. In addition to baseline fingerprint-based models, the work evaluates graph neural network encoders and a joint GNN+MLP fusion architecture for improved binding prediction. A RapidDock-inspired, distance-biased transformer branch is included as an exploratory prototype for structure-aware modeling.

Results show substantial performance improvements over early baselines, successful probability calibration for classification tasks, and coherent PK-PD trends in population-level simulations. The thesis concludes with practical limitations, reproducibility notes, and future directions toward external validation and deployment-oriented decision support.

Keywords: drug discovery, PK-PD, Neural ODE, multi-task learning, graph neural networks, calibration, dose optimization

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1. Introduction

Drug discovery pipelines face a fundamental trade-off between throughput and biological realism. Early-stage virtual screening requires fast computational models, while clinically relevant decision-making requires mechanistic understanding of pharmacokinetic and pharmacodynamic behavior. This thesis addresses this gap by combining data-driven molecular property prediction with ordinary differential equation (ODE)-based PK-PD simulation.

The project goal is to build a reproducible, end-to-end framework that starts from molecular representations and ends with decision-support outputs such as risk-benefit profiles, dose trade-offs, uncertainty estimates, and in-silico trial operating characteristics. The work combines multi-task machine learning, graph-based modeling, probability calibration, and population simulation.

1.1. Research Problem and Motivation

The main problem considered in this thesis is how to improve early drug candidate assessment by integrating:

1. multi-endpoint molecular prediction,
2. mechanistic PK-PD simulation,
3. and decision-oriented downstream analytics.

The practical motivation is to move from isolated endpoint prediction to coherent therapeutic assessment under uncertainty.

1.2. Objectives and Contributions

The thesis has four principal objectives:

1. Build a high-quality, leakage-controlled multi-task dataset from heterogeneous public sources.

2. Develop predictive models for binding affinity, hERG risk, Caco-2 permeability, and clearance.

3. Integrate model outputs with Neural ODE-based PK-PD simulation for scenario and optimization studies.

4. Evaluate calibration, interpretability, and robustness with clinically motivated analyses.

The key contributions are:

1. A full experimental pipeline spanning feature engineering, model training, calibration, and simulation.

2. A joint GNN+MLP fusion model that improves binding performance over individual branches.

3. Population-level analyses including sensitivity, Pareto optimization, Monte Carlo validation, and trial simulation.

4. An exploratory RapidDock-inspired prototype branch with documented protocol and qualitative feasibility assessment.

1.3. Thesis Organization

The remainder of the thesis is organized as follows. Chapter 2 presents theoretical background. Chapter 3 reviews related work and defines the research gap. Chapter 4 formalizes the problem and metrics. Chapter 5 describes data curation and preprocessing. Chapter 6 presents methods and model architectures. Chapter 7 defines the experimental setup. Chapter 8 reports empirical results. Chapter 9 discusses implications and limitations. Chapter 10 concludes and outlines future work.

2. Background and Theoretical Foundations

2.1. ADME-Tox Endpoints in Early Drug Discovery

Absorption, distribution, metabolism, excretion, and toxicity (ADME-Tox) endpoints are central to candidate selection. In this thesis, the following endpoints are modeled:

1. binding affinity (regression),
2. hERG-related cardiotoxicity risk (classification),
3. Caco-2 permeability proxy (classification),
4. clearance (regression).

2.2. Multi-Task Learning

Multi-task learning uses shared representations across related tasks to improve generalization, especially under partial labels. A shared encoder with task-specific heads is used in this work to model common molecular factors while preserving task-specific outputs.

2.3. Graph Neural Networks for Molecular Modeling

Molecules are naturally represented as graphs, where atoms are nodes and bonds are edges. Message Passing Neural Networks (MPNNs) aggregate local structural information into latent embeddings that can be used for property prediction.

2.4. PK-PD Modeling with ODEs

A two-compartment PK model with an effect-site linkage is used to bridge molecular predictions and pharmacodynamic response. A simplified form is:

$$\frac{dC_c}{dt} = -\frac{CL}{V_c}C_c - k_{12}C_c + k_{21}C_t + u(t), \tag{2.1}$$

$$\frac{dC_t}{dt} = k_{12}C_c - k_{21}C_t, \tag{2.2}$$

$$E(t) = E_{\max} \cdot \frac{C_e(t)^\gamma}{EC_{50}^\gamma + C_e(t)^\gamma}. \tag{2.3}$$

2.5. Calibration and Decision Reliability

For classification tasks, probability calibration is required when outputs are used in downstream risk computations. This thesis evaluates Expected Calibration Error (ECE) and applies temperature scaling and Platt scaling.

3. Related Work and Research Gap

Prior work in molecular machine learning has shown strong performance for single-endpoint prediction with fingerprints, graph networks, and transformers. In parallel, PK-PD literature has established mechanistic ODE-based models for dose-response analysis. However, integrated pipelines that combine robust multi-task molecular prediction, calibrated uncertainty, and decision-focused PK-PD simulation remain relatively limited.

The specific gap addressed here is the lack of a reproducible framework connecting molecular-level predictions to simulation-based therapeutic decision support in a single workflow.

4. Problem Formulation, Objectives, and Scope

4.1. Task Definitions

Let x denote molecular input features. The target vector is:

$$y = [y_{\text{bind}},\ y_{\text{hERG}},\ y_{\text{Caco2}},\ y_{\text{CL}}]\,. \tag{4.1}$$

Tasks are defined as:

1. $f_{\text{bind}}(x)$: regression of binding affinity,
2. $f_{\text{hERG}}(x)$: classification probability of hERG risk,
3. $f_{\text{Caco2}}(x)$: classification probability for permeability class,
4. $f_{\text{CL}}(x)$: regression of clearance.

4.2. Evaluation Metrics

Regression tasks are evaluated with R^2 and RMSE. Classification tasks are evaluated with AUROC, F1, and accuracy, with calibrated probability quality measured by ECE.

4.3. Scope and Constraints

The thesis focuses on data-driven preclinical decision support. It does not claim prospective clinical validation. External validation and deployment-grade clinical calibration are part of future work.

5. Data Sources, Curation, and Preprocessing

5.1. Data Sources

The workflow integrates public datasets including ChEMBL, TDC, ToxCast, PubChem, and PK-related sources. Data harmonization includes endpoint mapping, standardization of identifiers, and removal of invalid or duplicate entries.

5.2. Feature Engineering

Feature sets include physicochemical descriptors and Morgan fingerprints. In selected experiments, an additional docking-derived bridge signal is appended. Separate representations are maintained for baseline MLP and graph-based models.

5.3. Quality Control and Split Strategy

Quality checks include:

1. NaN/Inf detection,
2. duplicate handling,
3. data leakage audits between train/validation/test sets,
4. and class balance diagnostics.

6. Methodology and Model Architecture

6.1. Baseline Multi-Task MLP

The baseline model uses a shared encoder with four task heads. This establishes the initial reference for subsequent architecture improvements.

6.2. Fine-Tuning Strategy

Two fine-tuning variants are evaluated: heads-only updates and selective encoder unfreezing. Threshold calibration is applied post-training for classification heads.

6.3. Graph Encoder and Fusion Architecture

An MPNN-based encoder is used for structure-aware binding prediction. A joint GNN+MLP fusion model concatenates graph and tabular latent representations before final regression.

6.4. Neural ODE PK-PD Integration

Predicted molecular endpoints are mapped to PK-PD parameters and used in ODE simulation to derive concentration-time and effect-time profiles. The simulator is then used for downstream decision analysis.

6.5. RapidDock-Inspired Alternative Branch

As an alternative method track, a RapidDock-inspired distance-biased transformer is developed as a prototype branch. The branch uses molecular graph-derived geometric features to construct pairwise distance-biased attention signals and predicts binding-related outputs in a

structure-aware manner.

For reproducibility, the prototype follows the same data partitioning policy as the main modeling pipeline (train/validation/test split with leakage checks and scaffold-aware diagnostics), and is trained under the same implementation framework (PyTorch-based optimization, early stopping, and fixed random seeds where feasible). The primary optimization target is binding regression with RMSE-oriented model selection on the validation split.

This branch is intentionally scoped as exploratory: hyperparameter search is limited, calibration and uncertainty procedures are not fully integrated, and the branch is therefore reported as a feasibility pathway rather than a production candidate that replaces the main MLP and GNN+MLP tracks.

7. Experimental Setup and Protocol

7.1. Environment and Reproducibility

Experiments are implemented in Python with PyTorch and RDKit-based preprocessing. Random seeds and deterministic split logic are used where feasible to support reproducibility.

7.2. Training Protocol

The training setup includes early stopping, learning-rate scheduling, and task-specific losses. Hyperparameter sweeps are conducted for selected settings (e.g., class weighting and hidden dimensions).

7.3. RapidDock Prototype Protocol

The RapidDock-inspired branch is executed as a constrained prototype study. To ensure comparability, it reuses the same split definitions and preprocessing governance used by the primary experiments. The prototype is evaluated on binding-focused performance and scaffold-aware behavior checks, while extensive hyperparameter sweeps and full multi-endpoint calibration are deferred to future work.

7.4. Experimental Sequence

The experimental program proceeds from baseline modeling to advanced analytics:

1. baseline and feature upgrades,
2. fine-tuning and calibration,
3. GNN/fusion experiments,

4. RapidDock-inspired prototype feasibility run,
5. PK-PD simulation,
6. scenario, sensitivity, and optimization studies,
7. Pareto and Monte Carlo validation,
8. clinical-trial style simulation,
9. interpretation analysis.

8. Results

8.1. Predictive Modeling Results

The baseline multi-task model is progressively improved by feature upgrades and architecture tuning. Fine-tuning increases classification discrimination, while the joint fusion model yields the strongest binding performance among tested variants.

8.2. Calibration Results

Temperature scaling achieves substantial ECE reductions on hERG and Caco-2 tasks, improving reliability for downstream risk computations.

8.3. GNN and Fusion Results

Graph-based modeling improves binding prediction relative to tabular-only baselines. The joint GNN+MLP fusion model provides additional gains, indicating complementary signal between structural and fingerprint descriptors.

8.4. RapidDock Prototype Outcomes

The RapidDock-inspired branch demonstrated architectural feasibility and produced coherent scaffold-sensitive predictions in exploratory runs. However, due to constrained tuning budget and partial integration with the full calibration/uncertainty stack, results are interpreted qualitatively rather than as a definitive benchmark against the primary GNN+MLP pipeline. Accordingly, this branch is treated as a validated research direction for future expansion instead of a production replacement.

8.5. PK-PD and Decision Analytics Results

Population simulation, sensitivity analysis, dose optimization, Pareto front analysis, Monte Carlo confidence estimation, and trial simulation provide a coherent decision-support layer above endpoint prediction.

8.6. Interpretability Results

Feature-level saliency and atom-level analyses are used to identify dominant molecular signals. Cross-task overlap patterns and branch-ablation analyses provide additional insight into model behavior.

9. Discussion

The results indicate that integrating data-driven predictors with mechanistic simulation supports richer decision-making than isolated endpoint models. The workflow demonstrates both predictive and translational value, especially when calibration and uncertainty-aware analyses are included.

Limitations include dependence on public data quality, endpoint heterogeneity, and assumptions embedded in simulation parameters. The RapidDock branch is promising but requires broader validation and tighter integration with the main pipeline before production use.

10. Conclusion and Future Work

This thesis presents a complete neural PK-PD framework that links molecular prediction, calibration, mechanistic simulation, and decision analysis in one reproducible workflow. The combined approach improves endpoint modeling and enables clinically motivated in-silico analyses.

Future work should focus on:

1. external and prospective validation,
2. stronger mechanistic priors and uncertainty quantification,
3. expanded structure-aware modeling (including the RapidDock branch),
4. and deployment as an interactive decision-support tool.

Appendix

This appendix should include:

1. complete hyperparameter grids,
2. full metric tables for all experiments,
3. additional figures (calibration, Pareto, trial operating characteristics),
4. data schema and preprocessing details,
5. and reproducibility checklist (software versions, seeds, hardware notes).

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List of symbols and abbreviations

ADME	Absorption, Distribution, Metabolism, Excretion
PK/PD	Pharmacokinetics / Pharmacodynamics
ODE	Ordinary Differential Equation
GNN	Graph Neural Network
MPNN	Message Passing Neural Network
AUROC	Area Under Receiver Operating Characteristic
ECE	Expected Calibration Error
RMSE	Root Mean Squared Error
IIV	Inter-individual Variability

List of appendices

1. Hyperparameter search grids and training configuration snapshots.
2. Extended performance tables and calibration outputs.
3. Supplementary simulation plots (population, Pareto, and trial analyses).
4. Reproducibility sheet: package versions, seed settings, and runtime notes.